

HW2: fundamental frequency (f_0) estimation

Due Friday Oct. 15, 2021

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In this homework, you are given a starter code for viewing the short-time Fourier transform (STFT) as a video. You will explore a few different settings, and then your task is to estimate the fundamental frequency of human singing voice.

1. (10%) Select an appropriate frame length of STFT so you can clearly see separation of the fundamental frequency component and other harmonics. Report the range of such appropriate frame lengths for the female and the male voice, respectively.
2. (10%) Change the window type from rectangular to Hann. What difference do you see? Would that affect the choice of frame length for separation of harmonics?
3. (10%) What difference do you see when setting `sw.emphasis = 1` vs. 0?
4. (10%) In `<mySpecgram.m>`, try showing `abs(S)` in `imagesc()` instead of `20*log10(abs(S))`. Describe the difference you see visually. Which one do you prefer?
5. (40%) Write your own algorithm to detect the fundamental frequency as it varies from one frame of the signal to the next. We call the result “pitch contour”. You may make assumptions about the reasonable range of fundamental frequency of human voice, such as 80-320 Hz for male and 150-600 Hz for female, respectively, and then look for the most prominent peak in DFT in that range. Please be aware that $A_4 = 440$ Hz (the A note above the central C on a piano). Plot your result of pitch contour for both the female and male voice examples, respectively.
6. (10%) continuing from above, discuss any difficulties you encountered in implementing and testing your algorithm.

For this homework, please turn in your code and a brief report.